UAV Detection using CNN on radar spectrogram images

Ajnur Bogucanin

International University of Sarajevo

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Dr. Ali Abd Almisreb

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Abstract

This project uses EfficientNet to detect whether an object is a UAV using a collection of spectrograms recorded by a micro-doppler radar. The goal of this project is to take two drone classes and one non-drone class and attempt to achieve zero confusion between drones and non-UAV objects. The assumption is that if it is possible to make a good model with a simple spectrogram detection CNN, it should be beneficial to pursue a similar detection project for defense purposes using more complex data and architecture.

Keywords: UAV Detection, micro-doppler, radar

UAV Detection using CNN on radar spectrogram images

One of the most fearsome weapons of recent history and conflicts that emerged in Europe and the Middle East is armed UAVs. To achieve precise detection, models should detect different types of drones and objects that might move in a similar manner and that are not a threat. This difference is important since the model can easily mistake two quadrocopters, which is not dangerous, but the model feeding data to the interceptors should never mistake a drone for an aeroplane, for example, to avoid fatal and unwanted collateral damage. By training the model on different types of quadrocopters, there is a substantially smaller risk of getting a false positive.

This project uses Phantom 4 Pro and Mavic Pro 2 as two UAV classes and a corner reflector as a non-UAV class. The theory is that under no circumstances should the model mistake a stationary object like a corner reflector for a UAV.

The chosen detection range is 20m since the Mavic Pro 2 in sport flight mode can achieve a speed of 20m/s, so we have, in the worst-case scenario, 1 second to react to the threat with 80% accuracy at 20m and achieve higher accuracy as the drone approaches closer.

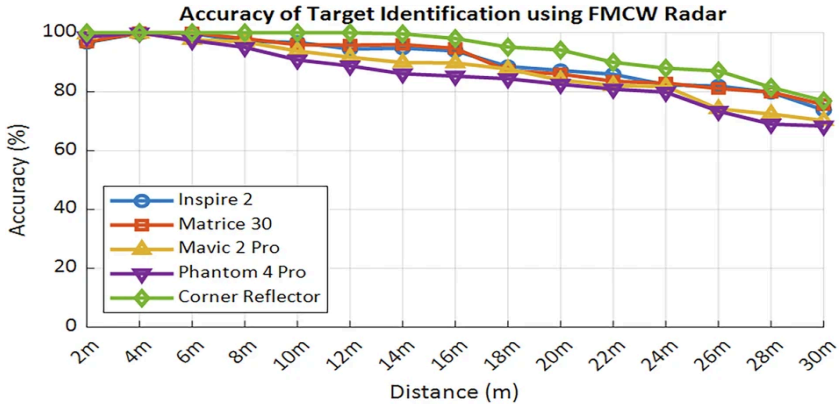


Figure 1.1: Range–Doppler representation of UAV radar signatures.

Adapted from Han & Jung (2026).

Dataset

This project utilizes the Time-Synchronized Multi-Sensor Drone Dataset (TSMS-Drone), a publicly available dataset designed for drone detection and classification research using multiple sensing modalities. The dataset was introduced by Seung-Kyu Han and Young-Ho Jung and contains synchronized measurements collected from three different sensor systems: a Continuous Wave (CW) radar, a Frequency-Modulated Continuous Wave (FMCW) radar, and a Radio Frequency (RF) receiver.

For this project, I used three out of five classes, using only images (spectrograms). The sensor system that was used in my project is FMCW. My derivative of this dataset features a total of 1500 images, 500 per class, with a split of:

training 350,

validation 75,

test 75.

Methodology

Tech Stack

The model is trained using Python and PyTorch. Dataset and transforms were done using torch and torchvision.

Architecure

The model that is trained for detection is EfficientNet, which is a CNN that is made for a balance between accuracy and compute. Resulting in a really fast training time and great results.

The architecture of the model can be seen in Figure 2.1 below.

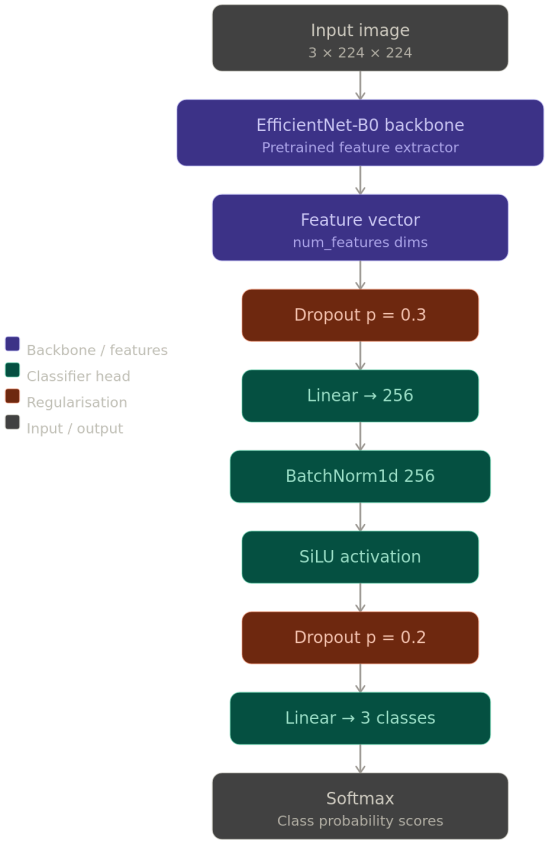


Figure 2.1: Model Architecture graph.

Input — RGB images resized to 224×224 pixels, the standard input resolution for EfficientNet.

EfficientNet-B0 Backbone — A lightweight convolutional neural network pretrained on ImageNet. Its final classification layer is removed, leaving a rich feature vector that encodes visual patterns from the input image.

Feature Vector — The backbone's output, a 1D tensor of length num_features, representing the image in a high-dimensional space.

Dropout (p=0.3) — Randomly zeroes 30% of features during training to reduce overfitting before the classification head.

Linear → 256 — A fully connected layer that projects the high-dimensional feature vector down to 256 dimensions, learning task-specific representations.

BatchNorm1d — Normalizes the 256-dimensional activations across the batch, stabilizing training and improving convergence speed.

SiLU Activation — Applies the Sigmoid Linear Unit non-linearity ($x \cdot \sigma(x)$), the same activation used internally by EfficientNet, maintaining architectural consistency.

Dropout (p=0.2) — A lighter second regularisation layer applied just before the final classifier, reducing co-adaptation of the learned features.

Linear → 3 — Maps the 256-dimensional representation to 3 raw logits, one per target class.

Softmax — Converts the logits into a probability distribution over the three classes, where all scores sum to 1.

Results

After 15 epochs, we got a model with the best performance of ~0.9911 accuracy. As an experiment, I made a UI program for testing accuracy on user-selected images. There was an occasional false positive, but the model performed really well most of the time.

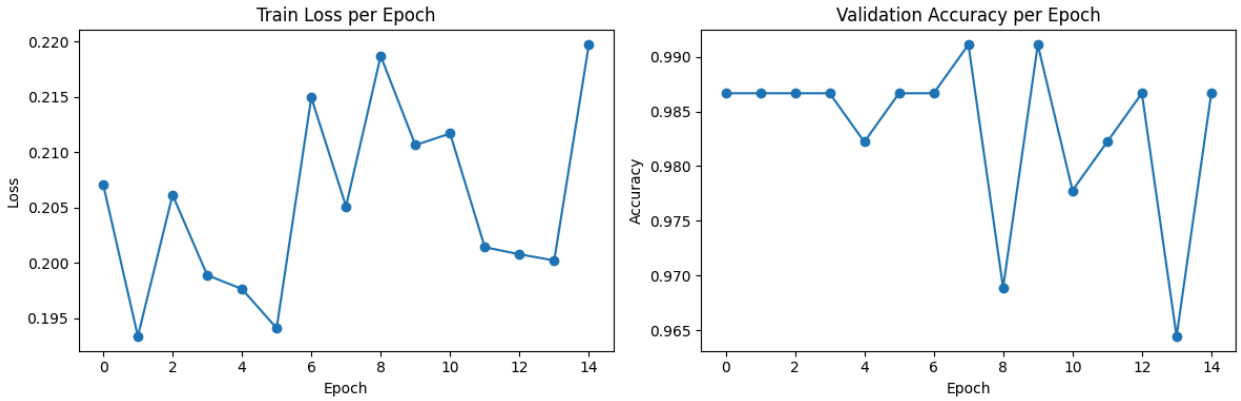


Figure 3.1: Model Loss and Accuracy graph.

We successfully achieved a smaller loss value during training. With confirmation of the practical model performance using a custom UI.

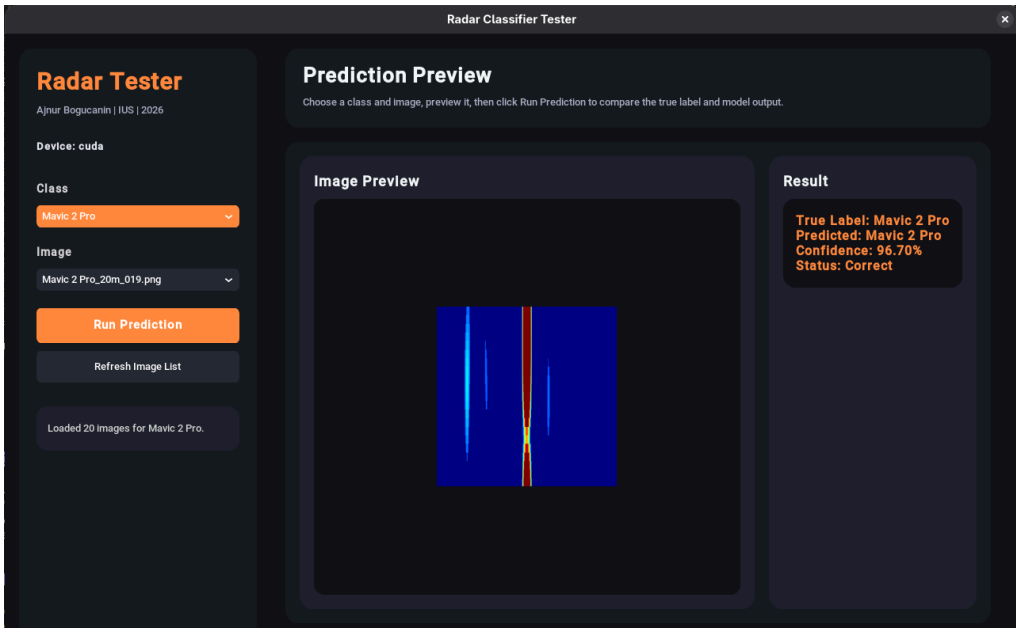


Figure 3.2: Radar Classification Tester program.

The most important premise that I have to confirm with this project to be able to pursue more complex projects, architectures, and systems in this direction, is that we would not get a single false positive with a substantially different class. If we can achieve this with an image detection model with a simple architecture, we should be able to achieve that using raw data and fused data from a radar, with more complex models.

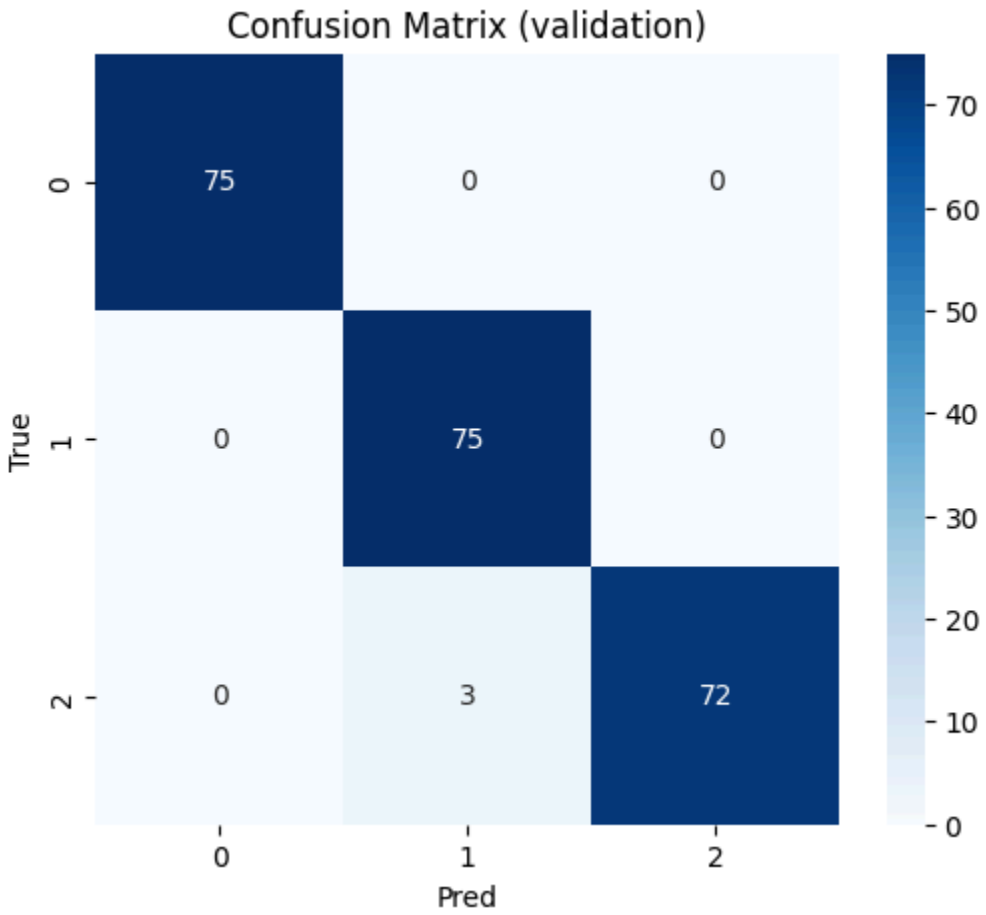


Figure 3.3: Confusion Matrix.

A confusion matrix shown in Figure 3.3 indicates that our model did not have a single false positive containing a label 0 (Corner reflector). This is really important since we want our model to be certain on simple classifications like such.

This system of converting raw signals to their visual representations made development easier for human interpretation, while also using that same data for model training. We can see in Figure 3.4 that there are visual cues that can be picked up by the naked eye, while we cannot say the same for the collection of data as tensors from raw data.

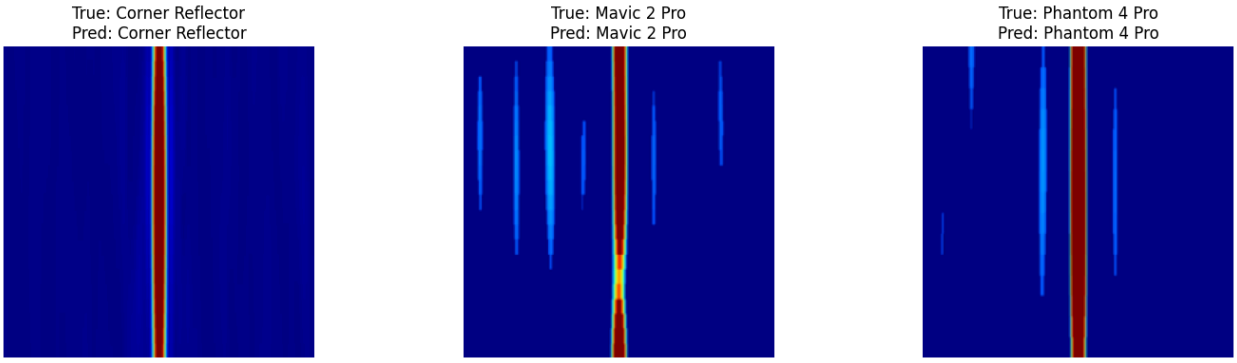


Figure 3.4 Prediction visual example.

Conclusion

The project achieved representative results with a low loss. Detection of UAVs is challenging but a task of great importance. This should be a good indicator that well-researched and carefully developed solutions in UAV detection could bear beneficial results, especially with a detection range greater than what we could ever expect from a camera. The goal of projects like this is to save lives, and that has to be done in time, before it is too late.

For the future, there should be stricter data leakage testing. This is a known problem when detecting from images like Doppler maps or spectrograms. Other great improvements would include the use of more modern and complex architectures and specialized data collection for defensive purposes.

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